

C389

Science, Technology, and Society

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A middle-school teacher who runs the district science fair tells her class that Newton's law of universal gravitation predicts the force between two masses, and asks what a theory of gravity would add. Her classroom has a new smartboard. What is the best answer?

- A. A more precise formula for the same force.
- B. An explanation of why masses attract, supported by many lines of evidence.
- C. A guess about gravity that will become a law once it is proven.
- D. Nothing, because a law is already absolute proof.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which statement is best described as a scientific law rather than a theory?

- A. A hypothesis that has been tested once and confirmed.
- B. At constant temperature, the pressure of a gas is inversely proportional to its volume.
- C. Species change over time through natural selection acting on variation.
- D. Matter is made of tiny particles in constant motion that explains gas behavior.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

On a field trip to a state park that opened in 1972, students see layers of rock with seashell fossils high on a canyon wall. The bus ride took two hours. Which statement is an inference?

- A. The canyon is two hours from the school.
- B. This area was once covered by an ocean.
- C. The rock shows several horizontal layers.
- D. There are shell-shaped fossils in the top layer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A student writes, “The plant is dying because it is not getting enough water.” Which statement is the observation that likely led to this inference?

- A.** The plant's leaves are brown and drooping.
- B.** The plant needs more water to survive.
- C.** All plants die without water.
- D.** The plant is probably getting too much sun instead.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A student at a school with a large vegetable garden tests whether the amount of fertilizer affects plant height. She gives four pots 0, 5, 10, and 15 grams of fertilizer, uses the same soil, water, and light, and measures height after three weeks. Her older sister won the science fair last year. What is the dependent variable?

- A.** Three weeks
- B.** Amount of fertilizer
- C.** Plant height
- D.** Soil, water, and light

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

In an experiment on whether light color affects how fast algae grow, what should be changed on purpose?

- A.** The type of container
- B.** The growth rate of the algae
- C.** Both light color and water temperature
- D.** The color of light

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

For decades, textbooks at a large publisher taught that stomach ulcers were caused mainly by stress. In the 1980s, researchers found strong evidence that a bacterium caused most ulcers, and treatment changed. The publisher employs over 500 people. What does this best show about scientific knowledge?

- A.** Scientific knowledge is reliable but can be revised when strong new evidence appears.
- B.** Textbook publishers decide what counts as science.
- C.** Once scientists agree, an idea is proven forever.
- D.** Science cannot be trusted because it keeps changing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A teacher with a background in journalism asks her class of 26 which question could be investigated scientifically. Which one qualifies?

- A.** Is it wrong to keep animals in zoos?
- B.** Do lucky charms work only when no one is watching?
- C.** Does the temperature of water affect how fast sugar dissolves?
- D.** Which color is the most beautiful?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A town with a historic main street needs a footbridge over a 6-meter creek. It must hold 20 people, cost under \$40,000, and be built in one summer. The mayor enjoys fishing in the creek. What is this task best described as?

- A.** A controlled scientific experiment
- B.** A problem with only one correct solution
- C.** Pure scientific research into creek ecology
- D.** An engineering design problem with criteria and constraints

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A student team built a water filter that met the cost limit but left the water cloudy. What should the team do next in the engineering design process?

- A.** Analyze why it failed, redesign, and test again against the criteria.
- B.** Report the design as final since it met the cost constraint.
- C.** Remove the cost limit so any design can pass.
- D.** Write a hypothesis and start over with a new question about nature.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A startup with a slick website issues a press release saying one study of 12 people proves its supplement doubles memory. The founder is a well-known television personality. What would most increase scientists' confidence in the claim?

- A.** A second press release with more detail
- B.** The founder's fame and media reach
- C.** Independent teams reproduce the result in larger, peer-reviewed studies.
- D.** The single study, since any published study is proof

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

In the 1600s, improved lenses made telescopes possible, and observers soon saw moons orbiting Jupiter. A later city council, facing a vote on water fluoridation, heard scientific testimony and public comment at a meeting that ran three hours. Which statement best describes these events?

- A.** Scientific data alone should decide the council's vote.
- B.** Technology only applies science and never drives new discoveries.
- C.** Science develops independently of technology and society.
- D.** Technology extended scientific observation, and society uses science to inform, but not alone decide, value-based policy.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A middle-school teacher ~~who runs the district science fair~~ tells her class that Newton's law of universal gravitation predicts the force between two masses, and asks what a theory of gravity would add. ~~Her classroom has a new smartboard.~~ What is the best answer?

Target word: what a theory adds to a law **Decides:** Laws describe; theories explain.

A	Confusion set	That would still be a description, the role of a law.
B	Correct	A theory explains the mechanism behind the pattern a law describes.
C	Inverted operation	Reverses the relationship; theories do not turn into laws.
D	Absolute language	Laws are well supported descriptions, not absolute proof, and they do not explain.

2. Answer B Concept 1 • transfer

Which statement is best described as a scientific law rather than a theory?

Target word: reverse: identify the law **Decides:** A law describes a regular pattern, often as an equation.

A	Dropped element	One confirmation makes neither a law nor a theory.
B	Correct	Describes a consistent, measurable relationship (Boyle's law).
C	Confusion set	That is an explanatory theory (evolution).
D	Adjacent category	Kinetic molecular theory explains the law's pattern.

3. Answer B Concept 2

On a field trip to a state park ~~that opened in 1972~~, students see layers of rock with seashell fossils high on a canyon wall. ~~The bus ride took two hours.~~ Which statement is an inference?

Target word: identify the inference **Decides:** A claim about the past drawn from present evidence.

A	Camouflage	Answers the travel detail; it is neither an inference about the rocks nor relevant.
B	Correct	A conclusion drawn from the fossils, not directly seen.
C	Confusion set	Directly seen: an observation.
D	Confusion set	Directly seen: an observation.

4. Answer A Concept 2 • transfer

A student writes, "The plant is dying because it is not getting enough water." Which statement is the observation that likely led to this inference?

Target word: reverse: find the observation behind an inference **Decides:** Observations are directly sensed.

A	Correct	Directly seen evidence that supports the inference.
B	Inverted operation	That is the inference itself, not the observation.
C	Absolute language	A general claim, not an observation of this plant.
D	Adjacent category	A competing inference, not an observation.

5. Answer C Concept 3

A student ~~at a school with a large vegetable garden~~ tests whether the amount of fertilizer affects plant height. She gives four pots 0, 5, 10, and 15 grams of fertilizer, uses the same soil, water, and light, and measures height after three weeks. ~~Her older sister won the science fair last year.~~ What is the dependent variable?

Target word: dependent variable **Decides:** What is measured as the outcome.

A	Wrong stem number	The time frame is a constant, not the outcome.
B	Inverted operation	That is the independent variable she changed.
C	Correct	Height is measured and depends on the fertilizer.
D	Confusion set	These are controlled variables held constant.

6. Answer D Concept 3 · transfer

In an experiment on whether light color affects how fast algae grow, what should be changed on purpose?

Target word: reverse: choose the independent variable **Decides:** The factor being tested is changed on purpose.

A	Adjacent category	A condition that should be held constant.
B	Inverted operation	That is measured: the dependent variable.
C	Dropped element	Changing two things at once confounds the result; other conditions must stay constant.
D	Correct	The independent variable is the factor tested.

7. Answer A Concept 4

For decades, textbooks ~~at a large publisher~~ taught that stomach ulcers were caused mainly by stress. In the 1980s, researchers found strong evidence that a bacterium caused most ulcers, and treatment changed. ~~The publisher employs over 500 people.~~ What does this best show about scientific knowledge?

Target word: what the ulcer case shows **Decides:** Strong new evidence revised an accepted idea.

A	Correct	Revision in response to evidence is how science self-corrects.
B	Camouflage	Seizes the publisher detail; evidence, not publishers, drove the change.
C	Absolute language	Consensus is strong but still open to evidence.
D	Inverted operation	Treats self-correction as a weakness rather than a strength.

8. Answer C Concept 5

A teacher ~~with a background in journalism~~ asks her class ~~of 26~~ which question could be investigated scientifically. Which one qualifies?

Target word: testable question **Decides:** Evidence could support or refute it.

A	Confusion set	A value question; evidence informs it but cannot settle it.
B	Dropped element	Built so no observation could test it.
C	Correct	Measurable variables that evidence could support or refute.
D	Adjacent category	A matter of taste, not testable.

9. Answer D Concept 6

A town ~~with a historic main street~~ needs a footbridge over a 6-meter creek. It must hold 20 people, cost under \$40,000, and be built in one summer. ~~The mayor enjoys fishing in the creek.~~ What is this task best described as?

Target word: science or engineering **Decides:** A problem with criteria and constraints to design for.

A	Confusion set	No question about nature is being tested; a solution is being designed.
B	Absolute language	Design problems allow many solutions that trade off criteria.
C	Camouflage	Answers the fishing detail, not the task.
D	Correct	Load is a criterion; budget and time are constraints; designs are tested and improved.

10. Answer A Concept 6 · transfer

A student team built a water filter that met the cost limit but left the water cloudy. What should the team do next in the engineering design process?

Target word: next step after a failed test **Decides:** Design is iterative.

A	Correct	Engineering iterates on designs using test data.
B	Dropped element	Drops the clarity criterion it failed.
C	Inverted operation	Changes the constraint instead of improving the design.
D	Adjacent category	Treats the design task as a science inquiry.

11. Answer C Concept 7

A startup ~~with a slick website~~ issues a press release saying one study of 12 people proves its supplement doubles memory. ~~The founder is a well-known television personality.~~ What would most increase scientists' confidence in the claim?

Target word: what builds scientific confidence **Decides:** Independent replication and peer review.

A	Adjacent category	Publicity is not peer review.
B	Camouflage	Reputation is not evidence.
C	Correct	Replication and review test whether the result holds.
D	Absolute language	One small study is not proof.

12. Answer D Concept 8

In the 1600s, improved lenses made telescopes possible, and observers soon saw moons orbiting Jupiter. A later city council, facing a vote on water fluoridation, heard scientific testimony and public comment ~~at a meeting that ran three hours~~. Which statement best describes these events?

Target word: how science, technology, and society interact **Decides:** Technology enabled observation; society weighs evidence with values.

A	Dropped element	Drops values, costs, and public input from a policy decision.
B	Inverted operation	Reverses the telescope case, where technology came first.
C	Absolute language	The telescope case shows technology driving discovery.
D	Correct	Both directions of the interaction plus the limits of science in policy.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Theory vs. law	A law describes a pattern in nature (often mathematically); a theory is a well-tested explanation of why it happens. Theories do not grow up into laws.	Believing a theory is a guess that becomes a law once proven.	"Explains why" vs. "describes what".
2	Observation vs. inference	An observation is gathered directly through the senses or instruments; an inference is a conclusion drawn from observations.	Treating an interpretation as if it were directly observed.	A claim about cause or history based on present evidence.
3	Variables in a controlled experiment	The independent variable is changed on purpose, the dependent variable is measured, and all other conditions are held constant.	Swapping independent and dependent, or changing two things at once.	"What did the student change?" vs. "what was measured?"
4	Tentative but durable knowledge	Scientific knowledge is reliable yet open to revision when strong new evidence appears.	Thinking science proves things with absolute, final certainty, or that revision means science is unreliable.	A long-accepted idea meets new evidence.
5	Testable questions	A scientific question can be answered by collecting evidence that could, in principle, show the idea is wrong.	Treating value, taste, or supernatural questions as scientific questions.	"Which question could be investigated scientifically?"
6	Engineering design vs. scientific inquiry	Science asks questions about the natural world; engineering defines a problem with criteria and constraints, then designs, tests, and iterates solutions.	Calling any hands-on building activity a scientific experiment, or skipping criteria and constraints.	A need, a budget, or a size limit in the stem.
7	Peer review and replication	Claims gain acceptance when methods are open, reviewed by other experts, and results are reproduced independently.	Accepting a claim because of the author's fame or a single striking result.	A dramatic single study or a press release.
8	Science, technology, and society interact	Technology extends what science can observe, science enables new technology, and society's values and funding shape both; science informs but cannot by itself settle value questions.	Thinking science develops in isolation, or that data alone settles ethical policy.	A new instrument, a funding decision, or a policy debate.

Course scope (public description): Science, Technology, and Society explores the ways in which science influences and is influenced by society and technology. A humanistic and social endeavor, science serves the needs of ever-changing societies by providing methods for observing, questioning, discovering, and communicating information about the physical and natural world. This course prepares educators to explain the nature and history of science, the various applications of science, and the scientific and engineering processes used to conduct investigations, make decisions, and solve problems. There are no prerequisites for this course.